Renode + Zephyr — Simulation Scope

What the S32K1–S32K3 bench can and cannot simulate without hardware · verified against current Renode source, July 2026

Capability	Status	Note
K1–K3 multi-node topology	**Simulable**	S32K388 + S32K118 merged; verified two-node demo
Unmodified firmware / Zephyr	**Simulable**	Same ELF as silicon; Zephyr tests pass
Healthy CAN + SocketCAN/vcan	**Simulable**	Frames reach host vcan0; Wireshark built in
Closed loop + reproducibility	**Simulable**	Virtual time; same seed → identical output
Memory / flash / arena footprint	**Simulable**	Structural, exact
Heap-exhaustion fault	**Simulable**	Analytical ground truth; no caveat
Fault injection (hooks)	**Workaround (yours)**	Standard hooks + your injection layer = Contribution 1
On-target latency (ms)	**Caveat**	Not cycle-accurate; indicative only, use board for hard numbers
Acoustic-anomaly data	**External**	Use MIMII dataset; inference runs on emulated core
CAN TEC/REC → bus-off	**Not in stock Renode**	Registers decorative, no error frames. Model counters in SW, extend FlexCAN in C#, or swap to timing signal